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Inventor(s)

KAGEYAMA; Masayuki

BATTERY

Abstract

A battery is provided and includes an electrode body that has a first surface with a positive electrode terminal and a negative electrode terminal located thereon; and an exterior member that houses the electrode body. The exterior member includes: a box member that has a box shape including a flat plate-shaped first side part that faces the first surface and has a through hole, a flat plate-shaped second side part located on the side opposite to the first side part, and a flat plate-shaped third side part that connects the first side part and the second side part, the box member being electrically connected to the negative electrode terminal; a plate member disposed inside the through hole to be separated from the first side part in a plan view from the first side part and electrically connected to the positive electrode terminal; and a sealing body overlapped with the first side part, the plate member, a gap between the first side part and the plate member in the plan view from the first side part, the sealing body sealing the gap.

Inventors: KAGEYAMA; Masayuki (Kyoto, JP)

Applicant: MURATA MANUFACTURING CO., LTD. (Kyoto, JP)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] The present application claims priority to Japanese patent application no. 2024-037148, filed on Mar. 11, 2024, the entire contents of which are incorporated herein by reference.

BACKGROUND

[0002] The present disclosure relates to a battery.

[0003] As an example of a battery, a rectangular battery is disclosed including a battery case with a cavity, a sealing plate for sealing the cavity, and an electrode plate group housed in the battery case. The electrode plate group has a positive electrode lead electrically connected to the sealing plate. The electrode plate group has a negative electrode lead electrically connected via an internal terminal to an external terminal penetrating the sealing plate, with the negative electrode lead electrically insulated from the sealing plate. The rectangular battery further includes a gasket and an insulator frame that electrically insulate the external terminal and the internal terminal from the sealing plate.

SUMMARY

[0004] The present disclosure relates to a battery.

[0005] In the rectangular battery described in the Background section, the internal terminal and the insulator frame are disposed inside the battery case in which the electrode plate group (electrode body) is housed. In this case, the volume of the electrode body is reduced with respect to the volume of the battery case, and the capacity of the battery is reduced. On the other hand, there is a desire for increasing the capacity of the battery.

[0006] The present disclosure, in an embodiment, relates to providing an increase in battery capacity.

[0007] A battery according to the present disclosure, in an embodiment, includes: an electrode body that has a first surface with a positive electrode terminal and a negative electrode terminal located; and an exterior member that houses the electrode body, in which the exterior member includes: a box member that has a box shape including a flat plate-shaped first side part that faces the first surface and has a through hole, a flat plate-shaped second side part located across the electrode body from the first side part, and a flat plate-shaped third side part that connects the first side part and the second side part, the box member being electrically connected to the negative electrode terminal; a plate member disposed inside the through hole to be separated from the first side part in a plan view from the first side part, and electrically connected to the positive electrode terminal; and a sealing body overlapped with the first side part, the plate member, and a gap between the first side part and the plate member in the plan view from the first side part, the sealing body sealing the gap.

[0008] The battery according to the present disclosure, in an embodiment, allows an increase in battery capacity to be achieved.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0009] FIG. 1 is a perspective view of a battery according to an embodiment of the present disclosure;

[0010] FIG. 2 is an exploded perspective view of the battery shown in FIG. 1;

[0011] FIG. 3 is a sectional view of the battery shown in FIG. 1;

[0012] FIG. 4 is a diagram illustrating a plan view of the battery viewed from a first side part;

[0013] FIG. 5 is an enlarged sectional view of a part of the sealing body shown in FIG. 3; and

[0014] FIG. 6 is an exploded perspective view of a battery according to a modification example of an embodiment of the present disclosure.

DETAILED DESCRIPTION

[0015] The present disclosure will be described in further detail including with reference to the drawings according to an embodiment. It is to be noted that the present disclosure is not limited by the embodiments. Each of the embodiments is illustrative, and obviously, parts of the configurations illustrated in the different embodiments can be replaced or combined with each other.

[0016] FIG. 1 is a perspective view of a battery 1 according to an embodiment of the present disclosure. FIG. 2 is an exploded perspective view of the battery 1 shown in FIG. 1. FIG. 3 is a sectional view of the battery 1 shown in FIG. 1.

[0017] The battery 1 is a secondary battery. The battery 1 is, for example, a lithium battery. The battery 1 includes an electrode body 10 and an exterior member 20.

[0018] The electrode body 10 is a wound-type electrode body. The electrode body 10 is formed by stacking and winding elongated positive electrode and negative electrode with a separator interposed therebetween. The electrode body 10 has a flattened shape. Hereinafter, the thickness direction, width direction, and depth direction of the electrode body 10 are referred to as a first direction D1, a second direction D2, and a third direction D3. The electrode body 10 has a first end surface 10a (corresponding to the “first surface”) and a second end surface 10b that face opposite sides in the third direction D3.

[0019] A first projected area of the electrode body 10 along the first direction D1 is larger than a second projected area of the electrode body 10 along the second direction D2. In addition, the second projected area is larger than a third projected area of the electrode body 10 along the third direction D3. The third projected area corresponds to the areas of the first end surface 10a and second end surface 10b. More specifically, the areas of the first end surface 10a and second end surface 10b are smaller than the first projected area and the second projected area.

[0020] The electrode body 10 includes a strip-shaped positive electrode terminal 11 electrically connected to the positive electrode and a strip-shaped negative electrode terminal 12 electrically connected to the negative electrode. The positive electrode terminal 11 and the negative electrode terminal 12 are located on the first end surface 10a of the electrode body 10.

[0021] The exterior member 20 houses the electrode body 10. The exterior member 20 includes a box member 30, a plate member 40, and a sealing body 50.

[0022] The box member 30 has a box shape in which the electrode body 10 is located. The box member 30 has conductivity. The material of the box member 30 is, for example, Fe (iron), Cu (copper), Ni (nickel), stainless steel, an iron alloy, a copper alloy, a nickel alloy, or the like. The type of stainless steel is not particularly limited, and specific examples thereof include SUS304 and SUS316.

[0023] The box member 30 has a first side part 31, a second side part 32, a third side part 33, a pair of fourth side parts 34, and a fifth side part 35.

[0024] The first side part 31 has a flat plate shape that faces the first end surface 10a of the electrode body 10 and has a through hole 31a. The second side part 32 has a flat plate shape located across the electrode body 10 from the first side part 31. The second side part 32 faces the second

end surface **10b**. The first side part **31** and the second side part **32** face each other with the electrode body **10** interposed therebetween in the third direction **D3**.

[0025] The third side part **33** has a flat plate shape that connects the first side part **31** and the second side part **32**. The third side part **33** faces the electrode body **10** in the first direction **D1**. The pair of fourth side parts **34** has a flat plate shapes that face each other with the electrode body **10** interposed therebetween in the second direction **D2**. The pair of fourth side parts **34** is connected by the first side part **31**, the second side part **32**, and the third side part **33** in the second direction **D2**. The fifth side part **35** has a flat plate shape located across the electrode body **10** from the third side part **33**. The third side part **33** and the fifth side part **35** face each other with the electrode body **10** interposed therebetween in the first direction **D1**.

[0026] In addition, the first side part **31**, the second side part **32**, the third side part **33**, and the pair of fourth side parts **34**, which are integrated, constitute a housing part **30a** that is open on one side in the first direction **D1**. In the housing part **30a**, the third side part **33** corresponds to a bottom plate, and the first side part **31**, the second side part **32**, and the pair of fourth side parts **34** correspond to side plates. Further, the flange part **F** is disposed on the peripheral edge of the cavity of the housing part **30a** over the whole perimeter.

[0027] The fifth side part **35** constitutes a lid **30b** that covers the cavity of the housing part **30a**. The peripheral edge of the lid **30b** is joined by, for example, laser welding to the flange part **F** over the whole perimeter. Thus, the housing part **30a** and the lid **30b** are electrically connected, and the housing part **30a** and the lid **30b** are sealed therebetween.

[0028] The negative electrode terminal **12** is joined by, for example, resistance welding to the inner surface of the housing part **30a** (for example, the inner surface of the fourth side part **34**). Thus, the housing part **30a** is electrically connected to the negative electrode terminal **12**. More specifically, the box member **30** is electrically connected to the negative electrode terminal **12**.

[0029] FIG. **4** is a diagram illustrating a plan view of the battery **1** viewed from the first side part **31**.

[0030] The plate member **40** shown in FIGS. **1**, **2**, **3**, and **4** has a flat plate shape and has conductivity. The thickness of the plate member **40** is substantially equal to the thickness of the first side part **31**. The material of the plate member **40** is aluminum, an aluminum alloy, or the like. In addition, the material of the plate member **40** may be, for example, a clad material obtained by rolling and joining: a metal layer made of one or more of Fe (iron), Cu (copper), Ni (nickel), stainless steel, an iron alloy, a copper alloy, and a nickel alloy; and an aluminum layer.

Furthermore, for example, a composite member that has a three-layer structure of an Al (aluminum) layer, a stainless steel layer, and a nickel layer may be applied to the plate member **40**. The type of stainless steel is not particularly limited, and for example, SUS304, SUS316, and the like.

[0031] The plate member **40** is disposed inside the through hole **31a** to be separated from the first side part **31** in a plan view from the first side part **31**. The plate member **40** faces the first end surface **10a** of the electrode body **10**. The plan view from the first side part **31** has a gap **G** between the first side part **31** and the plate member **40** over the whole perimeter of the peripheral edge of the plate member **40**. The plate member **40** is electrically insulated from the box member **30** by the gap **G**.

[0032] As shown in FIG. **3**, the positive electrode terminal **11** is joined by, for example, laser welding to the inner surface of the plate member **40**. Thus, the plate member **40** is electrically connected to the positive electrode terminal **11**.

[0033] In addition, the battery **1** further includes a first insulator **60** and a second insulator **70**. The first insulator **60**, which has a strip shape, is disposed on the positive electrode terminal **11**. The first insulator **60** electrically insulates the positive electrode terminal **11** from the exterior member **20**. The second insulator **70** is disposed on the first end surface **10a** of the electrode body **10**. The second insulator **70** electrically insulates the positive electrode terminal **11** from the negative electrode.

[0034] The sealing body **50** seals the gap **G** between the first side part **31** and the plate member **40**. The sealing body **50** has an annular shape in a plan view. The sealing body **50** is overlapped with the first side part **31**, the plate member **40**, and the gap **G** in a plan view from the first side part **31**. The sealing body **50** is disposed on the outer surface of the first side part **31**. The plate member **40** is exposed from the inside of the sealing body **50**.

[0035] FIG. **5** is an enlarged sectional view of a part of the sealing body **50** shown in FIG. **3**. The sealing body **50** includes a substrate layer **51**, a joining layer **52**, and a protective layer **53**.

[0036] The material of the substrate layer **51** is, for example, stainless steel. The type of stainless steel is not particularly limited, and specific examples thereof include SUS304 and SUS316. In addition, the material of the substrate layer **51** may be, for example, a simple metal such as aluminum, iron, copper, and nickel, and alloys thereof, that is, an aluminum alloy, an iron alloy, a copper alloy, a nickel alloy, and the like.

[0037] The joining layer **52** is overlapped with the substrate layer **51** to join the first side part **31** and the plate member **40** to the substrate layer **51**. The joining layer **52** has an electrical insulation property. The material of the joining layer **52** is a thermoplastic resin. The material of the joining layer **52** is, for example, polypropylene (PP), polyethylene (PE), or the like. In addition, the material of the joining layer **52** may be unstretched polypropylene (CPP), polyphenylene sulfide (PPS), or the like.

[0038] The protective layer **53** is overlapped with the substrate layer **51** on the side opposite to the joining layer **52** to protect the substrate layer **51**. The protective layer **53** has an electrical insulation property. The heat resistance of the protective layer **53** is higher than the heat resistance of the joining layer **52**. The material of the protective layer **53** is a thermoplastic resin. The material of the protective layer **53** is, for example, nylon, polyethylene terephthalate (PET), or the like.

[0039] The sealing body **50** is joined by thermal welding to the first side part **31** and the plate member **40**. The joint strength between the sealing body **50** and the first side part **31** and plate member **40** is smaller than the joint strength between the housing part **30a** and the lid **30b** in the box member **30**.

[0040] In addition, when the internal pressure of the exterior member **20** is equal to or higher than a predetermined pressure, the sealing body **50** is cleaved. More specifically, the sealing body **50** functions as a cleavage valve. The predetermined pressure is a pressure at which the safety of battery **1** can be ensured.

[0041] For example, when the positive electrode and the negative electrode are short-circuited in the electrode body **10**, gas is generated from the electrode body **10**, and the electrode body **10** abnormally generates heat. Thus, the joining layer **52** is softened, and when the internal pressure of the exterior member **20** reaches the predetermined pressure or higher, the joining layer **52** is cleaved, and the gas leaks from the gap **G**. Accordingly, the internal pressure of the exterior member **20** is kept from being increased, and the safety of the battery **1** is secured.

[0042] The respective materials and thicknesses of the substrate layer **51**, joining layer **52**, and protective layer **53** are determined such that the sealing body **50** is cleaved when the internal pressure of the exterior member **20** is equal to or higher than the predetermined pressure.

[0043] As described above, inside the box member **30**, the third side part **33** and the fifth side part **35** face each other with the electrode body **10** interposed therebetween in the first direction **D1**, the pair of fourth side parts **34** face each other with the electrode body **10** interposed therebetween in the second direction **D2**, and the first side part **31** and the second side part **32** face each other with the electrode body **10** interposed therebetween in the third direction **D3**. In addition, the positive electrode terminal **11** and the negative electrode terminal **12** are both disposed on the first end surface **10a**, and thus, the first side part **31** faces the first end surface **10a** of the electrode body **10** where the positive electrode terminal **11** and the negative electrode terminal **12** are located.

[0044] When the positive electrode terminal **11** and the negative electrode terminal **12** are both disposed on the first end surface **10a**, the gap between the second side part **32** and the electrode

body **10** can be reduced, for example, as compared with a case where the positive electrode terminal **11** is disposed on the first end surface **10a**, whereas the negative electrode terminal **12** is disposed on the second end surface **10b**. Similarly, when the positive electrode terminal **11** and the negative electrode terminal **12** are both disposed on the first end surface **10a**, the gap between the third side part **33**, pair of fourth side parts **34**, and fifth side part **35** and the electrode body **10** can be reduced. Accordingly, the volume of the electrode body **10** can be increased with respect to the volume of the exterior member **20**.

[0045] In addition, the plate member **40** has a flat plate shape that has a thickness substantially equal to the thickness of the first side part **31**, and is located inside the through hole **31a** of the first side part **31**. More specifically, the plate member **40** is kept from being located inside the box member **30**. Furthermore, the sealing body **50** is disposed on the outer surface of the first side part **31**. Thus, the thickness of the member disposed on the first side part **31** in the third direction **D3** can be reduced. Thus, the gap between the first side part **31** and the electrode body **10** can be reduced. Accordingly, the volume of the electrode body **10** can be increased with respect to the volume of the exterior member **20**. More specifically, the capacity of the battery **1** can be increased.

[0046] In addition, as mentioned above, in the electrode body **10**, the area of the first end surface **10a** corresponding to the projected area in the third direction is smaller than the first projected area in the first direction **D1** and the second projected area in the second direction **D2**. Thus, when the positive electrode terminal **11** and the negative electrode terminal **12** are located on the first end surface **10a**, the gap between the electrode body **10** and the box member **30** can be reduced as compared with a case where the positive electrode terminal **11** and the negative electrode terminal **12** are located on a site of the electrode body **10** facing the third side part **33** or the fourth side part **34**. Accordingly, the capacity of the battery **1** can be further increased.

[0047] In addition, as mentioned above, the sealing body **50** functions as a cleavage valve. More specifically, the battery **1** does not need to separately include any cleavage valve. Thus, for the battery **1**, the number of parts can be reduced, the manufacturing process can be reduced, and the cost can be reduced.

[0048] It is to be noted that the embodiments mentioned above are intended to facilitate understanding of the present disclosure, but not intended to construe the present disclosure in any limited way. The present disclosure may be modified or improved without departing from the spirit of the present disclosure, and the present disclosure includes equivalents of the present disclosure.

[0049] For example, the electrode body **10** may be a rectangular parallelepiped stacked body that has a plurality of sheet-shaped positive electrodes and a plurality of sheet-shaped negative electrodes alternately stacked with separators interposed therebetween.

[0050] In addition, the housing part **30a** may have no flange part **F**. In this case, the peripheral edge of the lid **30b** is joined to the peripheral edge of the cavity of the housing part **30a**.

[0051] FIG. **6** is an exploded perspective view of the battery **1** according to a modification example of the embodiment of the present disclosure. The battery **1** according to the present modification example differs from the battery **1** according to the embodiment mentioned above in the configuration of a box member **130**. For the box member **130** according to the present modification example, a second side part **132**, a third side part **133**, a pair of fourth side parts **134**, and a fifth side part **135**, which are integrated, constitute a housing part **130a** that is open on one side in the third direction **D3**. In addition, a first side part **131** constitutes a lid **130b** that covers the cavity of the housing part **130a**. The peripheral edge of the lid **130b** is joined to the peripheral edge of the cavity of the housing part **130a** over the whole perimeter. Further, in the present modification example, the pair of fourth side parts **134** may have arc sectional shape.

[0052] It is to be noted that the present disclosure may be a combination of the following configurations according to an embodiment. [0053] (1)

[0054] A battery including: [0055] an electrode body that has a first surface with a positive electrode terminal and a negative electrode terminal located; and [0056] an exterior member that

houses the electrode body, [0057] in which the exterior member includes: [0058] a box member that has a box shape including a flat plate-shaped first side part that faces the first surface and has a through hole, a flat plate-shaped second side part located across the electrode body from the first side part, and a flat plate-shaped third side part that connects the first side part and the second side part, the box member being electrically connected to the negative electrode terminal; [0059] a plate member disposed inside the through hole to be separated from the first side part in a plan view from the first side part, and electrically connected to the positive electrode terminal; and [0060] a sealing body overlapped with the first side part, the plate member, and a gap between the first side part and the plate member in the plan view from the first side part, the sealing body sealing the gap. [0061] (2)

[0062] The battery according to (1), in which [0063] the sealing body includes: [0064] a substrate layer; and [0065] a joining layer overlapped with the substrate layer to join the first side part and the plate member to the substrate layer. [0066] (3)

[0067] The battery according to (2), in which a material of the joining layer is a thermoplastic resin. [0068] (4)

[0069] The battery according to any one of (1) to (3), in which the sealing body is disposed on an outer surface of the first side part. [0070] (5)

[0071] The battery according to any one of (1) to (4), in which when an internal pressure of the exterior member is equal to or higher than a predetermined pressure, the sealing body is cleaved.

[0072] It should be understood that various changes and modifications to the embodiments described herein will be apparent to those skilled in the art. Such changes and modifications can be made without departing from the spirit and scope of the present subject matter and without diminishing its intended advantages. It is therefore intended that such changes and modifications be covered by the appended claims.

Claims

1. A battery comprising: an electrode body that has a first surface with a positive electrode terminal and a negative electrode terminal located; and an exterior member that houses the electrode body, wherein the exterior member includes: a box member that has a box shape including a flat plate-shaped first side part that faces the first surface and has a through hole, a flat plate-shaped second side part located across the electrode body from the first side part, and a flat plate-shaped third side part that connects the first side part and the second side part, the box member being electrically connected to the negative electrode terminal; a plate member disposed inside the through hole to be separated from the first side part in a plan view from the first side part, and electrically connected to the positive electrode terminal; and a sealing body overlapped with the first side part, the plate member, and a gap between the first side part and the plate member in the plan view from the first side part, the sealing body sealing the gap.

2. The battery according to claim 1, wherein the sealing body includes: a substrate layer; and a joining layer overlapped with the substrate layer to join the first side part and the plate member to the substrate layer.

3. The battery according to claim 2, wherein a material of the joining layer includes a thermoplastic resin.

4. The battery according to claim 1, wherein the sealing body is disposed on an outer surface of the first side part.

5. The battery according to claim 1, wherein when an internal pressure of the exterior member is equal to or higher than a predetermined pressure, the sealing body is cleaved.
